

Vasilios Nicholas

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Available: January 2027 – December 2027 | GitHub: github.com/vasiliosnicholas | Personal Site: vasilios-nicholas.com

EDUCATION

Northeastern University – Boston, MA Aug 2025 – Present
MS in Computer Science Candidate, GPA: 4.0 / 4.0 *Expected Graduation: December 2027*
Relevant Coursework: Algorithms; Object-Oriented Design; Data Structures, Algorithms and their Applications within Computer Systems;
Web Development; Master's Research; Computer Systems (in progress);

Johns Hopkins University – Baltimore, MD Aug 2020 – Aug 2021
MS in Healthcare Management Aug 2016 – May 2020
BS in Neuroscience, Degree Conferred with General Honors

TECHNICAL KNOWLEDGE

Languages: Python, Java, C, C++, JavaScript, TypeScript, HTML5, CSS3, Bash, LaTeX, Markdown

Tools, Frameworks, Systems: Git, GDB, NumPy, Pandas, SciPy, Matplotlib, JUnit, React.js, Svelte, SvelteKit, Node.js, Express.js, MongoDB, Docker, Podman, Unix/Linux, Microsoft Office, Claude Code, GitHub, Bitbucket, VS Code, JetBrains IDE's (IntelliJ, PyCharm), VIM, UML

PROJECTS

Pupillometry Filtering Package | Python, TypeScript, Svelte | *Northeastern SATH Lab, Massachusetts General Hospital* Jan 2026 – Present
• Developing an open-source pupillometry noise filtering and analysis package as a Khoury Master's Research Apprentice and [Research Assistant](#).
• [Presented](#) the blink detection and filtering algorithm at the Khoury Master's Research Symposium on April 14th, 2026
• Currently developing a web app for researchers to clean their pupillometry data using the package.

Agility — Agile Project Management Platform | TypeScript, React, Node.js, Express, MongoDB | *Northeastern University* July – Aug 2026
• Built the authentication and team-management stack of an Agile project management application, deployed on Render with MongoDB Atlas.
• Implemented session-based authentication with Passport Local and bcrypt, enforcing role-based route guards across the API and SPA.
• Developed the team management, ticket assignment, and profile CRUD flows with React Hook Form and Yup; added arrow-key navigation.

NavHealth — Healthcare Facility Search Engine | JavaScript (ES6), Node.js, Express, MongoDB | *Northeastern University* June – July 2026
• Built the reviews subsystem of a search application that surfaces healthcare facilities by service, insurance accepted, and distance.
• Implemented a **REST API** with full CRUD over a MongoDB reviews collection, sorted by recency, rating, or popularity.
• Rendered review interfaces client-side with vanilla JavaScript fetch calls, over a database seeded from Massachusetts facility profile datasets.

CS 5004 Object-Oriented Design — Adventure Game | Java | *Northeastern University* Nov – Dec 2025
• Designed and implemented a text- and GUI-based (**Swing**) adventure game using **Model View Controller** architecture and **SOLID** principles.
• Wrote **JUnit** tests covering the model and controller modules to verify each component's functionality.
• Modeled the class hierarchy in **Unified Modeling Language (UML)** before implementation, designing for readability and extensibility, including using a single controller to handle both the text- and GUI-based view modules.

WORK EXPERIENCE

Research Assistant Intern | Massachusetts General Hospital – Boston, MA May 2026 – Present
• Developing an **open-source Python** package that detects and filters blink artifacts from raw pupillometry recordings, enabling collaborating clinical researchers to evaluate the kinematics of pupil diameter as a biomarker of circadian system health.
• Implemented a **blink-filtering algorithm** and validated output against manually labeled data to confirm physiological signal is preserved.
• **Scaled** the algorithm to clean **840+** study signals reducing total subsequent manual cleaning time **to under 2 hours**.
• **Evaluated linear and non-linear regression models** using goodness-of-fit metrics to identify the optimal model for quantifying pupil kinematics.
• Designed the **API and documentation** for clinical researchers, eliminating a critical, time-intensive data-cleaning bottleneck.
• Developing a **front-end app** for researchers to clean and view analysis on their pupillometry data using the package in **Svelte and TypeScript**.

Graduate Teaching Assistant | Northeastern University, Khoury College – Boston, MA Jan 2026 – Apr 2026
• Led a weekly recitation and biweekly office hours for a **DSA** course for graduate students entering CS from non-technical backgrounds.
• Used recurring errors identified during grading to restructure recitation content, targeting the conceptual gaps the class had rather than those assumed by the syllabus.
• Developed comprehensive **C test suites** for each assignment to expose edge cases in students' **DSA implementations** and provide actionable feedback.
• Graded assignments on correctness, algorithmic complexity, and code readability, giving written feedback to **20 students weekly**.

Clinical Research Coordinator | The Institute for Advanced Reconstruction – Tinton Falls, NJ Jun 2022 – Aug 2023
• Analyzed clinical datasets for prospective and retrospective studies, computing descriptive and inferential statistics that supported 3 manuscripts and conference abstracts.
• Maintained **IRB compliance** across 2 active clinical studies, managing protocol documentation, consent records, and data integrity; trained an incoming coordinator on the same workflows.
• **Authored** literature reviews and evidence-based insurance appeals that reversed coverage denials for reconstructive surgery patients.

Data Analyst | Seaview Orthopaedics and Medical Associates – Ocean Township, NJ Mar 2021 – Oct 2021
• Collected and analyzed patient engagement for the practice to identify bottlenecks in scheduling and follow-up workflows.
• Partnered with administration and IT to evaluate, select, and deploy new patient engagement resulting in increased patient portal adoption.

LEADERSHIP AND VOLUNTEERING

Holmdel First Aid Squad – Holmdel, NJ | Volunteer EMT-B and Crew Chief Jan 2019 – Aug 2023
• **Led** crews of 2-4 across approximately **400 annual 911 responses**, assessing, treating, and transporting patients, escalating to advanced life support when required, assisting with motor vehicle extrication, and assigning crew roles and assessing crew performance under time pressure.

HONORS, AWARDS, AND CERTIFICATIONS

Honors: Spring 2026 Khoury Master's Research Apprenticeship (Northeastern University), Dean's List (Johns Hopkins University, Fall 2016, Spring 2017, Fall 2019); Nu Rho Psi National Neuroscience Honor Society (2019); Eagle Scout (2016)

Certifications: NJ Emergency Medical Technician (EMS ID 638304, expired); AHA Basic Life Support Provider; PEARS Provider

Languages: English (fluent); Greek (intermediate), German (intermediate)